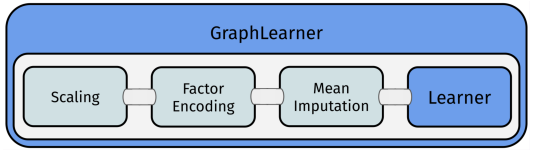
# Dataflow programming with mlr3pipelines::CHEAT SHEET

## Introduction

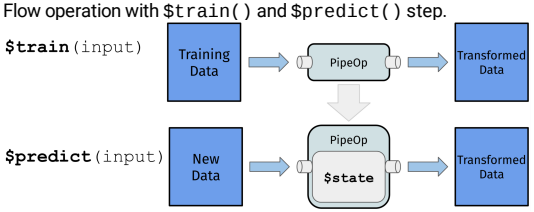
Combine ML operations to flexible pipelines and processing graphs, which can be configured trained, resampled, tuned as any regular learner. The main purpose of a `Graph` is to build combined preprocessing and model fitting pipelines that can be used as a `Learner`.




Each operation in the above example is a `PipeOp` which transforms the data in each step. `PipeOps` are chained with the %>>% operator.

## PipeOp

Flow operation with `$train()` and `$predict()` step.




Construction example: `pca = po("pca")`

- $train(input): Named list
- $predict(input): Named list
- $state: Learned parameters
- $param_set: See hyperparameters

## Popular PipeOps

| Class | Key | Operation |
|---|---|---|
| PipeOpRemoveConstants | "removeconstants" | Repair Tasks |
| PipeOpScale | "scale" | Scale Features |
| PipeOpImputeMean | "impute" | Impute NAs |
| PipeOpFilter | "filter" | Feature Filter |
| PipeOpEncode | "encode" | Factor Encoding |
| PipeOpPCA | "pca" | PCA |
| PipeOpSelect | "select" | Restrict Columns |
| PipeOpColApply | "colapply" | Transform Columns |
| PipeOpClassBalancing | "classbalancing" | Imbalanced Data |
| PipeOpLearner | "learner" | Use Learner |
| PipeOpLearnerCV | "learner_cv" | Crossval Learner |
| PipeOpMutate | "mutate" | Fearure Engineering |
| PipeOpChunk | "chunk" | Split Data |
| PipeOpSubsample | "subsample" | Subsample Rows |
| PipeOpFeatureUnion | "featureunion" | Combine Features |
| PipeOpFixFactors | "fixfactors" | Handle Unknown Levels |
| PipeOpNOP | "nop" | Do Nothing |

Full list: `as.data.table(mlr_pipeops)`

## Graph

Connects `PipeOps` with edges to control data flow during training and prediction. Input is sent to sources (no in-edges), output is read from sinks (no out-edges).
Important methods and slots:

- Display: `print(gr)`, `gr$plot(html = TRUE)`
- Accessing `PipeOps`: `gr$pipeops`
  Named list of all contained POs.

## Graph Construction

The %>>% operator takes either a `PipeOp` or a `Graph` on each of its sides and connects all left-hand outputs to the right-hand inputs.
For full control, connect `PipeOps` explicitly:

```
gr = Graph$new()
gr$add_pipeop(po("pca"))
gr$add_pipeop(lrn("classif.rpart"))
gr$add_edge("pca", "classif.rpart")
```

## GraphLearner

`GraphLearner` behave like `Learner` and enable all mlr3 features : `grl = GraphLearner$new(gr)` See slots `$encapsulatefor debugging and$model` for results after training.

## Linear Graphs

Concatenate POs with %>>%:

### Example

```
gr = po("scale") %>>% po("encode") %>>%
  po("imputemean") %>>% lrn("classif.rpart")
grl = GraphLearner$new(gr)
# access the scale pipeop:
grl$graph$pipeops$scale
grl$train(task)
grl$model
grl$predict(task)
rr = resample(task, grl, rsmp("cv", folds = 3))
```

## Hyperparameters

For POs: Exactly as in a `Learner`.

```
enc = po("encode")
enc$param_set
enc$param_set$values = list(method="one-hot")
po("encode", param_vals = list(method="one-hot"))
```

For `Graph` / `GraphLearner`: All HPs are collected in a global ParamSet stored in `$param_set`. IDs are prefixed with the respective `PipeOp`'s `id`.

## Tuning

Can jointly tune any Pipeline.

### Example

```
gr = po("encode") %>>% lrn("classif.rpart")
grl = GraphLearner$new(gr)
tune_ps = ParamSet$new(list(
  ParamFct$new("encode.method",
    levels = c("one-hot", "poly")),
  ParamDbl$new("classif.rpart.cp",
    lower = 0, upper = 0.05)
))
tt = trm("evals", n_evals = 20)
rs = rsmp("holdout")
inst = TuningInstanceSingleCrit$new(task, grl, rs,
  msr("classif.ce"), tune_ps, tt)
tuner = tnr("random_search")
tuner$optimize(inst)
```

Usage of `AutoTuner` is identical.

## Feature Engineering

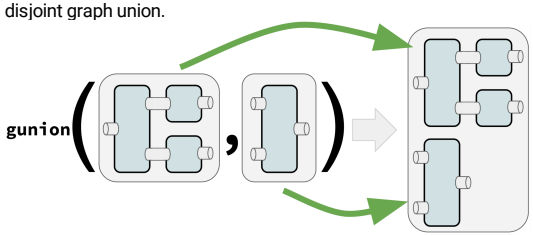
`PipeOpMutate` adds new features. This works by providing expressions in a `list`.

### Example

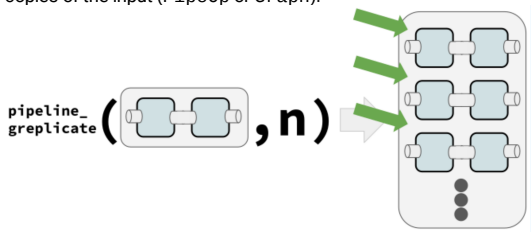
```
task = tsk("iris")
mutations = list(
  Sepal.Sum = ~ Sepal.Length + Sepal.Width)
mutate = po("mutate", param_vals =
  list(mutation = mutations))
GraphLearner$new(mutate %>>% lrn("classif.rpart"))
```

## Nonlinear Graphs

`gunion()arrangesPipeOps orGraph`s next to each other in a disjoint graph union.




`pipeline_greplicate()` creates a new `Graph` containing n copies of the input (`PipeOp` or `Graph`).




`PipeOpFeatureUnion` aggregates features from all input tasks into a single `Task`.

### Example

```
# train on orig and pca features
gunion(list(po("nop"), po("pca"))) %>>%
  po("featureunion") %>>% lrn("classif.rpart")
```

### Example

```
pr = po("subsample") %>>% lrn("classif.rpart")
bagging = ppl("greplicate", pr, n = 10) %>>%
  po("classifavg", innum = 10)
```

## Branching

Controls the path execution. Only one branch can be active. Which one is controlled by a hyperparameter. Unbranching ends the forking.

### Example

```
gr = ppl("branch", list(
  pca = po("pca"), scale = po("scale"))
)
# set the "pca" path as the active one:
gr$param_set$values$branch.selection = "pca"`
```

Tuning the branching selection enables powerful model selection.